

Lab 02 - First Verilog

In this lab, you've learned how to do an initial and simple design in Verilog to learn the Vivado tooling and process involved in RTL/FPGA design.

Rubric

Item	Description	Value
Summary Answers	Your writings about what you learned in this lab.	25%
Question 1	Your answers to the question	25%
Question 2	Your answers to the question	25%
Question 3	Your answers to the question	25%

Lab Summary

This lab was an introduction to verilog design and how to set up a project on vivado as well as initializing your program to the board of your choosing. This lab was pretty straightforward and the only difficult part was getting our files to simulate properly.

Lab Questions

1 - Describe the stages of building a Verilog project in Vivado.

- First you create a project repository then you initialize which type of project it is then copy over the files from your system or create files and ensure its synthesis or simulation or both, add a constraint file and add your board. Then just simulate your project and synthesize it to the board you added and program it.

2 - What is the value in looking at the elaborated design schematic?

- The value in looking at a elaborated design schematic is that you can see which switches and leds are initialized and if they providing input/output

3 - Why should we simulate our designs frequently? What does the simulation do?

- It allows us to see what changes have been made to the switches input and output as well as saving the code that we have added to the files. The simulation shows us which 'sw' and 'led' we initialized.

Code Submission

Upload a .zip of all your code or a public repository on GitHub.